

A • FORCE — Definition, Types & Effects (Cl. 6 & 8)

Force (Class 6)

A push or pull by an animate or inanimate agency that causes or tends to cause motion, stop motion, change direction, or change shape of an object.

Force (Class 8)

An external agency which changes or tends to change the state of rest or uniform motion of a body, its direction, or its shape. Force is a **vector quantity**.

SI Unit

Newton (N) (CGS unit = dyne ; 1 N = 10⁵ dyne)

Force =

F = m × a (Newton's 2nd Law; m in kg, a in m/s²)

Weight =

W = m × g (g = 9.8 m/s² ≈ 10 m/s² on Earth's surface)

Contact & Non-Contact Forces (Class 6)

Type	Acts when...	Class 6 Examples	Class 8 Addition
Contact Force	Physical touch required	Kicking ball, pulling cart, wind on grass	Applied force, Normal force, Tension, Friction
Non-contact Force	No physical contact needed	Gravity (coconut falls), Magnet attracts iron	Electrostatic force, Magnetic force
Muscular Force	Muscles exert force (contact)	Lifting bag, swimming, cycling	Type of contact force

Effects of Force (Class 8 — Section 2.1.1)

Effect of Force	Explanation	Example
Set object in motion	Stationary body begins to move	Kick a ball at rest
Stop moving object	Moving body brought to rest	Braking a bicycle
Change speed	Velocity increases or decreases	Accelerating car
Change direction	Direction of moving object changes	Batsman deflects ball
Change shape/size	Object compressed, stretched, or deformed	Squeezing balloon, pulling rubber band
Effect ∝ Force magnitude	Greater force → greater effect	Hard bat-hit → ball goes farther

■ FORCE — TET MCQ Quick-Check

Q: What is the SI unit of force?

Ans: Newton (N)

Q: 1 Newton = ? dyne

Ans: 10⁵ dyne

Q: Force is a _____ quantity

Ans: Vector (has magnitude + direction)

Q: Gravity is an example of _____ force

Ans: Non-contact force

Q: Formula: Force = ?

Ans: $F = m \times a$ (mass $\times$ acceleration)

B · MOTION — Rest, Types & Kinematics (Cl. 6 & 7)

Motion (Class 6)

When there is a **change in position** of an object with respect to time and its surroundings, the object is said to be in motion.

Rest (Class 6)

An object is at **rest** when its position does NOT change with respect to its surroundings. *Note: Rest and Motion are RELATIVE — not absolute.*

Types of Motion Based on Path (Class 6 — Section 2.2)

Type	Description	Class 6 Examples	Additional Examples
Linear (Rectilinear)	Straight-line path	Person walking on straight road; Falling stone	Car on highway
Curvilinear	Curved path (moving but changing dir.)	Thrown paper aeroplane	Ball thrown at angle
Circular	Circular path around a fixed centre	Swirling stone on rope	Earth around Sun, fan blade tip
Rotatory	Spinning on its own axis	Spinning top; sharpening pencil in sharpener	Earth's rotation; motor
Oscillatory	To-and-fro about a mean position	Pendulum; pencil held between fingers	Swing; vibrating string
Zigzag (Irregular)	Random directions; no fixed path	Fly buzzing in room	People in crowded street

Periodic vs Non-Periodic Motion (Class 6)

Periodic Motion — repeats after a fixed time interval.

- Oscillatory motion → always periodic (pendulum)
- Circular motion → periodic (Earth's revolution)
- Rotatory motion → periodic (Earth's rotation)

Non-Periodic Motion — no fixed time pattern.

- Zigzag / Irregular motion (fly, crowd)
- Curvilinear motion (thrown ball)
- Linear motion may be non-periodic

Distance & Displacement (Class 7 — Section 2.1)

	Distance	Displacement
Definition	Total length of path travelled from start to end	Shortest straight-line path from start to end point
Type	Scalar (no direction)	Vector (has direction)
SI Unit	metre (m)	metre (m)
Can it be zero?	Only if object never moved	Yes — if object returns to start (circular trip)
Equal when?	—	When object moves in a straight line without turning
Symbol	d	s (or Δx)

Nautical Mile

1 nautical mile = 1.852 km — used in aviation and sea transport. **Knot** = 1 nautical mile/hour — speed unit for ships and aircraft.

C · Speed, Velocity & Acceleration (Cl. 6 & 7)

Speed (Class 6 — Section 2.3 & Class 7 — Section 2.2.1)

Speed

$$s = d / t \text{ (Distance } \div \text{ Time | scalar quantity | SI unit: m/s)}$$

Average Speed

$$v_{\text{avg}} = \text{Total Distance} / \text{Total Time} \text{ (Used when speed varies)}$$

Conversion

$$1 \text{ km/h} = 5/18 \text{ m/s} \mid 1 \text{ m/s} = 18/5 \text{ km/h} = 3.6 \text{ km/h}$$

Speed Type	Definition	Class	Example
Uniform Speed	Equal distances in equal time intervals	6 & 7	Light through vacuum (3×10^8 m/s)
Non-uniform Speed	Unequal distances in equal time intervals	6 & 7	Bus in city traffic
Average Speed	Total distance $\div$ total time taken	7	Total trip with varying speeds

Velocity (Class 7 — Section 2.2.2)

Velocity

$$v = \text{Displacement} / \text{Time} \text{ (vector | SI unit: m/s)}$$

Average Velocity

$$v_{\text{avg}} = \text{Total Displacement} / \text{Total Time} \text{ (positive/negative based on direction)}$$

Velocity Type	Definition	Example
Uniform Velocity	Equal displacement in equal time in same direction	Light through vacuum; ideal uniform circular motion
Non-uniform Velocity	Speed OR direction changes (or both)	Train leaving station; vehicle turning corner
Average Velocity	Total displacement $\div$ total time taken	Car going east 5 km then west 7 km

C.2 · Acceleration — Types (Class 7, Section 2.3)

Acceleration

$$a = (v - u) / t \text{ (v=final velocity, u=initial velocity, t=time | SI unit: m/s}^2\text{)}$$

Acceleration Type	Definition (Class 7)	Sign	Example
Positive Acceleration	Velocity INCREASES with time (Section 2.3.1)	+ve	Car speeding up; Cheetah sprinting 0 $\rightarrow$ 20 m/s in 2 s
Negative Acceleration (Deceleration / Retardation)	Velocity DECREASES with time (Section 2.3.2)	–ve	Golf ball slowing down; braking bus
Uniform Acceleration	Change in velocity is SAME per unit time (Section 2.3.3)	+ve/–ve	Bus velocity increasing 20 m/s every second
Non-uniform Acceleration	Change in velocity is DIFFERENT per unit time (Section 2.3.4)	Varies	City traffic vehicle; velocity 0,10,40,60,70,50 m/s

■ MOTION — TET MCQ Quick-Check

Q: Cheetah speed = 25–30 m/s. It goes 0 $\rightarrow$ 20 m/s in 2 s. Acceleration = ?

Ans: 10 m/s²

Q: What is the SI unit of acceleration?

Ans: m/s² (metre per second squared)

Q: Displacement = 0 means...

Ans: Object returned to starting point

Q: 1 nautical mile = ?

Ans: 1.852 km

Q: Speed is a _____ quantity; velocity is a _____ quantity

Ans: Scalar; Vector

D · Distance–Time & Speed–Time Graphs (Class 7, Sections 2.4 & 2.5)

Distance–Time Graph (Section 2.4) — 4 Cases

Graph Shape / Slope	What it Means	Motion Type
Horizontal line (zero slope)	Distance constant — object NOT moving	Rest
Straight line, constant slope	Distance increases uniformly — constant speed	Uniform speed
Curve with INCREASING slope	Distance increases faster — speed is rising	Acceleration
Curve with DECREASING slope	Distance increase slowing — speed is falling	Deceleration

Speed–Time Graph (Section 2.5) — 6 Cases

Graph Shape	What it Means	Motion Type
Horizontal line at speed = 0	Bus not moving	Rest (zero speed)
Horizontal line at speed > 0	Constant speed, zero acceleration	Uniform speed
Straight line RISING (const. slope)	Speed increasing at constant rate	Uniform acceleration
Straight line FALLING (const. slope)	Speed decreasing at constant rate	Uniform deceleration
Rising curve with INCREASING slope	Acceleration itself is increasing	Non-uniform acceleration (increasing)
Rising curve with DECREASING slope	Acceleration itself is decreasing	Non-uniform acceleration (decreasing)

D–T vs S–T Graph Comparison (Section 2.5.1)

Journey Phase	Distance–Time Graph	Speed–Time Graph
Accelerating uniformly from rest	Concave curve (gradient increasing)	Straight line rising (constant positive slope)
Constant speed	Straight line (constant gradient)	Horizontal line (zero slope = zero acceleration)
Decelerating uniformly to stop	Convex curve (gradient decreasing)	Straight line falling (constant negative slope)
Area under graph	Not directly useful	= Distance travelled

E · Centre of Gravity & Stability (Class 7, Sections 2.6 & 2.7)

Centre of Gravity (CG)

The single point through which the **entire weight of an object appears to act**. For regular-shaped objects, CG lies at the geometric centre.

Shape	Location of Centre of Gravity
Square / Rectangle	Intersection of diagonals (geometric centre)
Circle / Disc	Centre of the circle
Triangle	Centroid (intersection of medians)
Ring / Hollow circle	Centre of the ring (may be in empty space)

Shape	Location of Centre of Gravity
Uniform ruler	Midpoint of the ruler

Types of Equilibrium / Stability (Class 7 — Section 2.7)

Type	When Displaced...	CG change	Returns?	Example
Stable Equilibrium	CG rises briefly; weight-line stays WITHIN base	CG rises	YES ✓	Frustum resting on wide base; Thanjavur doll; Racing car
Unstable Equilibrium	CG falls; weight-line falls OUTSIDE base	CG lowers	NO ✗	Inverted frustum balanced on tip; pencil balanced on tip
Neutral Equilibrium	CG stays at SAME height; object rolls but does not topple	No change	Same position	Frustum on its side (rolls); sphere on flat surface

Increasing Stability — Conditions (Section 2.7.1)

Stability **INCREASES** by:

1. **Lowering the Centre of Gravity** (heavy base)
2. **Increasing the base area** (broad base)

Application examples (Class 7):

- Tour bus luggage at **bottom** (not on roof)
- Racing cars built **low and broad**
- Table lamps have **large heavy base**
- Extra passengers **NOT** on upper deck of double-decker bus

Thanjavur Doll — Stable Equilibrium:

- Made of terracotta material
- CG and total weight concentrated at **bottommost point**
- Even when tilted, returns to upright position
- Produces slow oscillatory dance-like movement

Finding CG of irregular shapes:

Suspend shape from 3 different holes; draw plumb-line each time.

Intersection of three lines = Centre of Gravity.

F · Pressure — Thrust, Air, Liquid & Pascal's Law (Class 8)

Thrust	Force acting perpendicularly on a surface — unit: Newton (N). A large force over a small area creates very high pressure.
Pressure	Amount of thrust (force) acting on unit area of a surface. $P = F/A$. SI unit: Pascal (Pa) = Nm^{-2} — named after Blaise Pascal.

Pressure $P = F / A$ (Thrust ÷ Area) (SI: Pascal (Pa) = 1 Nm^{-2} | CGS: dyne/cm²)

Pressure — Practical Examples (Class 8)

Situation	Area	Pressure	Why?
Sharp knife cuts easily	Small (sharp edge)	HIGH	Same force, tiny area → huge pressure → penetrates easily
Injection needle pierces skin	Very small tip	HIGH	Concentrated force on tiny area
Camel walks on sand	Large padded feet	LOW	Large area distributes weight → low pressure → no sinking
Heavy truck → many wheels	Large (many tyres)	LOW	More wheels = more contact area = less road pressure
Broad bag straps	Wide strap	LOW	Wider strap spreads force over shoulder → comfortable
Nail hammered into wall	Sharp tip	HIGH	Small tip area creates very high pressure → penetrates

Atmospheric Pressure (Class 8 — Section 2.3)

Parameter	Value / Fact
Definition	Weight of the entire column of air above unit area of Earth's surface
Standard value	1 atm = 76 cm Hg = 760 mmHg = 1,01,325 Pa $\approx 1.01 \times 10^5 \text{ Nm}^{-2}$
Measurement instrument	Barometer (mercury barometer) — invented by Evangelista Torricelli (1643)
Pressure vs Altitude	Atmospheric pressure DECREASES as altitude increases above Earth
Why tilting barometer doesn't change reading	Level of mercury column remains the same regardless of tilt angle
Cooking at high altitude	Lower atm. pressure $\rightarrow$ boiling point decreases (water boils even at 80°C) $\rightarrow$ cooking difficult
Pressure cooker principle	Sealed vessel $\rightarrow$ increased pressure inside $\rightarrow$ boiling point rises $\rightarrow$ food cooks faster

Liquid Pressure (Class 8 — Section 2.4.1)

Liquid Pressure $P = \rho \times g \times h$ (ρ =density (kg/m ³), g =9.8 m/s ² , h =depth (m))	
Property of Liquid Pressure	Explanation
Increases with depth	Pressure at bottom of container > pressure at top (spouting can experiment)
Same in all directions at same depth	Liquid exerts equal pressure sideways, up, down at a given depth
Depends on height of liquid column	Not on shape or volume of container — only height (h) matters
Buoyant Force	Upward force on submerged/floating object = weight of liquid displaced (Archimedes)
Object floats if...	Weight of object < Buoyant force (upward force)
Object sinks if...	Weight of object > Buoyant force
Why dams wider at base?	Water pressure increases with depth $\rightarrow$ greater force at bottom $\rightarrow$ needs thicker wall
Why scuba divers wear suits?	To withstand high water pressure at great ocean depths

Pascal's Law (Class 8 — Section 2.4.2)

Pascal's Law		
Pressure applied at any point of a liquid at rest in a closed system is distributed equally through all directions of the liquid. (Blaise Pascal, 1648)		
Application	Working Principle	Where Used
Hydraulic Lift / Jack	Small force on small piston $\rightarrow$ same pressure $\rightarrow$ large force on large piston ($F_1/A_1 = F_2/A_2$)	Car service stations, elevators
Hydraulic Brakes	Brake fluid transmits pressure from pedal equally to all four wheel brake cylinders	Automobiles (bikes, cars, trucks)
Hydraulic Press	High pressure applied to compress materials into smaller volume	Cotton/paper bale compression
Syringe / Pump	Pressure on liquid transmitted to outlet nozzle	Medical injections, pumps

G · Surface Tension & Viscosity (Class 8, Sections 2.5 & 2.6)

Surface Tension	Property of liquid surface molecules to contract and minimize surface area. Defined as force per unit length on the liquid surface. Unit: Nm⁻¹
Viscosity / Viscous Force	Frictional force between successive layers of a moving liquid that opposes their relative motion. Unit: poise (CGS) or kgm⁻¹s⁻¹ = Nsm⁻² (SI)

Property	Surface Tension	Viscosity
Acts on	Surface of liquid (boundary layer)	Between moving liquid layers (throughout)
Unit	Nm^{-1}	Poise (CGS) $\text{kgm}^{-1}\text{s}^{-1}$ (SI)
Effect of temperature	Decreases when temperature rises	DECREASES when temperature rises (honey flows faster when hot)
Practical use	Capillary action in plants, drops become spherical	Engine oil lubrication, blood flow, ink flow
Reduced by	Adding detergent/soap → spreads better → cleans	Heating the liquid; adding lubricants

Applications of Surface Tension (Class 8 — Section 2.5.1)

- **Rain drops spherical** — liquid contracts to minimum surface area
- **Capillary action** in xylem tissue of plants — water rises against gravity
- **Water strider insect** walks on water surface
- **Paper clip floats** on water surface (heavier than water)
- **Needle floats** on carefully placed on water
- **Detergent lowers surface tension** → water spreads into cloth fibers → better cleaning
- **Sailors pour oil on rough sea** to reduce surface tension and calm waves
- **Soap bubbles** form because of liquid film surface tension
- **Mosquito larvae** breathe through the water surface (surface tension supports them)

H · Friction — Types, Factors, Pros, Cons & Control (Class 8)

Friction

Force arising between two or more bodies in contact that **opposes relative motion**. Caused by geometrical dissimilarities (roughness) of surfaces. Acts opposite to direction of motion.

Types of Friction (Class 8 — Section 2.7.1)

Type	When it Occurs	Magnitude	Class 8 Example
Static Friction	Body at REST; external force applied but no motion yet	LARGEST (max. just before motion)	Book resting on table, parked vehicle
Sliding Friction (Kinetic)	One surface SLIDES over another during motion	MODERATE	Box dragged on floor; chalk on blackboard; braking car
Rolling Friction	Body ROLLS over a surface	SMALLEST (least)	Ball rolling on ground; bicycle wheel; car tyre
Fluid Friction (Viscosity)	Object moving through liquid or gas	Depends on speed & viscosity	Ship in water; bird in air; swimming

Order of Friction **Rolling < Sliding < Static** (That is why ball bearings (rolling) used in machines to reduce friction)

Factors Affecting Friction (Class 8 — Section 2.7.2)

Factor	Effect on Friction	Example
Nature of surface (rough vs smooth)	Rough surface → MORE friction; Smooth surface → LESS friction	Sand road vs polished marble floor
Weight / Mass of body	Greater weight → GREATER friction	Loaded trolley harder to push than empty one
Area of contact	For given weight: greater area → greater friction	Road roller (broad base) vs cycle tyre (narrow)

Advantages (2.7.3) vs Disadvantages (2.7.4) of Friction

Advantages of Friction ■	Disadvantages of Friction ■
We can walk without slipping	Wears out surfaces — soles, tyres, gears, machine parts
Vehicles brake and stop safely	Wastes energy — extra effort needed to overcome friction
We write with pen/pencil on paper	Produces HEAT — damages machines and causes wear
Knots, nails, bolts, screws hold tight	Noise produced in machines by friction
Matchstick lights when struck	Vehicles slow down due to air/road friction → fuel wasted
Climbing, gripping, stitching possible	Moving parts need frequent maintenance due to friction

Friction is a "Necessary Evil"

It is necessary (we cannot live without it) yet evil (it wastes energy and causes wear). Therefore called a **necessary evil**.

Increasing & Decreasing Friction (Class 8 — Section 2.7.5)

Method	Increases or Decreases?	How it Works	Example
Increasing area of contact	INCREASES friction	More surface → more contact irregularities	Brake shoes adjusted close to wheel rim
Lubricants (oil/grease)	DECREASES friction	Fills gaps in rough surfaces; smooth layer prevents direct contact	Engine oil, coconut oil, castor oil, grease, graphite
Ball Bearings	DECREASES friction	Converts sliding friction to rolling friction (rolling < sliding)	Cycle hub, electric motors, axles, wheels
Streamlining	DECREASES fluid friction	Smooth body shape reduces air/water resistance	Aircraft, ships, sports cars, fish body shape
Polishing surfaces	DECREASES friction	Removes surface irregularities → smoother contact	Sports tracks, machine parts
Using rough / grooved surface	INCREASES friction	More grip required for safety	Tyre treads, brake pads, shoe soles, stair mats

■ FRICTION — TET MCQ Quick-Check

Q: Which friction is LEAST?

Ans: Rolling friction (smallest of all three types)

Q: Why are ball bearings used in cycle hub?

Ans: To convert sliding friction to rolling friction (rolling < sliding)

Q: What is the unit of viscosity in SI system?

Ans: $\text{kg m}^{-1} \text{s}^{-1}$ (= N s m^{-2})

Q: Friction is called a "necessary evil" because...

Ans: It is essential for daily life BUT wastes energy and causes wear

Q: Which factor does NOT affect friction?

Ans: Color / material color of the surfaces

I · Solved Numerical Problems

Problem 1	A vehicle covers 400 km in 5 hours. Calculate its speed.	Problem 2	A car accelerates from rest to 20 m/s in 10 seconds. Find acceleration.
Given	Distance $d = 400 \text{ km} = 4,00,000 \text{ m}$; Time $t = 5 \text{ h} = 18,000 \text{ min}$	Given	Initial velocity $u = 0 \text{ m/s}$; Final velocity $v = 20 \text{ m/s}$; Time $t = 10 \text{ s}$
Solution	Speed $= d/t = 4,00,000 / 18,000 = 22.22 \text{ m/s}$ OR in $\text{km/h} = 80 \text{ km/h}$	Solution	$a = (v - u) / t = (20 - 0) / 10 = 2 \text{ m/s}^2$
Answer	80 km/h = 22.22 m/s	Answer	Acceleration = 2 m/s²

Problem 3	A golf ball decelerates from 8 m/s to 2 m/s in 10 s. Find deceleration.	Problem 4	An elephant weighs 4000 N. Area of one foot sole = 0.1 m ² . Find pressure.
Given	$u = 8 \text{ m/s}$; $v = 2 \text{ m/s}$; $t = 10 \text{ s}$	Given	Force on one foot $= 4000 \div 4 = 1000 \text{ N}$; Area $A = 0.1 \text{ m}^2$
Solution	$a = (v - u) / t = (2 - 8) / 10 = -6/10 = -0.6 \text{ m/s}^2$	Solution	$P = F/A = 1000 / 0.1 = 10,000 \text{ Nm}^{-2}$
Answer	Deceleration = 0.6 m/s² (retardation)	Answer	10,000 Pa = 10 Nm⁻²

Problem 5	A stone weighs 500 N, contact area = 25 cm ² . Find pressure.	Problem 6	Geetha cycles at 2 m/s for 15 minutes. Find distance. (Class 7)
Given	$F = 500 \text{ N}$; $A = 25 \text{ cm}^2 = 25 \times 10^{-4} \text{ m}^2 = 0.0025 \text{ m}^2$	Given	Speed $v = 2 \text{ m/s}$; Time $t = 15 \text{ min} = 900 \text{ s}$
Solution	$P = F/A = 500 / 0.0025 = 2,00,000 \text{ Nm}^{-2}$	Solution	Distance $= v \times t = 2 \times 900 = 1800 \text{ m}$
Answer	$2 \times 10^5 \text{ Pa}$	Answer	1800 m = 1.8 km

J · KEY FACTS — Rapid Revision (Numbers, Years & Names)

Formulas, Values & Units — Memorise These

Quantity / Concept	Formula / Value	Unit / Remark
Force (F)	$F = m \times a$	Newton (N) CGS: dyne
1 Newton	$= 10^5 \text{ dyne}$	Unit conversion
Weight (W)	$W = m \times g$ ($g = 9.8 \text{ m/s}^2 \approx 10$)	Newton (N)
Speed (s)	$s = d/t$ (distance $\div$ time)	m/s km/h
1 km/h	$= 5/18 \text{ m/s}$ (multiply by 5/18 to convert)	Unit conversion
1 m/s	$= 18/5 \text{ km/h} = 3.6 \text{ km/h}$	Unit conversion
Velocity (v)	$v = \text{displacement} \div \text{time}$	m/s (vector)
Acceleration (a)	$a = (v - u) / t$	m/s ²
Pressure (P)	$P = F/A = \text{Thrust} / \text{Area}$	Pascal (Pa) = Nm ⁻²
1 atm	$= 76 \text{ cm Hg} = 760 \text{ mmHg} = 1.01 \times 10^5 \text{ Pa}$	Standard value
Liquid pressure	$P = \rho \times g \times h$ (density $\times g \times$ depth)	Pa (Nm ⁻²)
Surface tension	Force per unit length on liquid surface	Nm ⁻¹
Viscosity unit (CGS)	Poise	CGS unit
Viscosity unit (SI)	$\text{kg m}^{-1} \text{ s}^{-1} = \text{N s m}^{-2}$	SI unit
1 Nautical mile	$= 1.852 \text{ km}$	Sea/air navigation
Cheetah top speed	25–30 m/s ($0 \rightarrow 20 \text{ m/s}$ in $2 \text{ s} = 10 \text{ m/s}^2$ acc.)	Class 7 textbook data
Speed of light	$3 \times 10^8 \text{ m/s}$ (vacuum)	Uniform velocity example

Scientists, Discoverers & Years

Scientist / Person	Contribution	Year / Class
Isaac Newton	Laws of Motion; Law of Gravitation; Unit "newton" named after him	1687 Cl. 8

Scientist / Person	Contribution	Year / Class
Blaise Pascal	Pascal's Law (fluid pressure transmission); Unit "pascal" named after him	1648 Cl. 8
Evangelista Torricelli	Invented mercury barometer — measures atmospheric pressure	1643 Cl. 8
Galileo Galilei	Law of falling bodies; studied free fall, velocity, acceleration	1638 Cl. 7
Archimedes	Archimedes' Principle: buoyant force = weight of displaced fluid	287–212 BC Cl. 8
Aryabhata	Observed that rest and motion are RELATIVE (river bank appears moving)	5th century CE Cl. 6

K · GLOSSARY — Definitions

Force — Push or pull by animate/inanimate agency; SI unit: Newton	Thrust — Force acting perpendicularly on a given surface area
Pressure — Thrust per unit area ($P = F/A$); SI unit: Pascal (Pa)	Pascal (Pa) — SI unit of pressure; $1 \text{ Pa} = 1 \text{ Nm}^{-2}$; named after Blaise Pascal
Newton (N) — SI unit of force; $1 \text{ N} =$ force that accelerates 1 kg mass at 1 m/s^2	Motion — Change in position of an object with respect to time and a reference
Rest — No change in position with respect to time and reference point	Distance — Total length of path travelled — scalar quantity
Displacement — Shortest straight-line path from start to end, with direction — vector	Speed — Rate of change of distance — scalar quantity; unit: m/s
Velocity — Rate of change of displacement — vector quantity; unit: m/s	Acceleration — Rate of change of velocity; $a = (v-u)/t$; unit: m/s^2
Deceleration — Negative acceleration; velocity decreasing with time (retardation)	Linear Motion — Motion along a straight-line path
Circular Motion — Motion along a circular path around a fixed centre	Rotatory Motion — Spinning of a body about its own axis
Oscillatory Motion — To-and-fro motion of a body about a mean (rest) position	Curvilinear Motion — Motion along a curved path where direction constantly changes
Periodic Motion — Motion that repeats itself after a fixed time interval	Uniform Motion — Equal distances covered in equal time intervals
Atmospheric Pressure — Weight of atmospheric air per unit area of Earth's surface	Barometer — Instrument to measure atmospheric pressure (mercury barometer)
Buoyant Force — Upward force exerted by a fluid on a floating or submerged object	Pascal's Law — Pressure applied to a confined liquid is transmitted equally in all directions
Hydraulic Lift — Device using Pascal's Law; small force on small piston → large force on large piston	Surface Tension — Property of liquid surface to contract; defined as force per unit length; unit: Nm^{-1}
Viscosity — Property of a liquid to resist flow; internal friction between moving liquid layers	Viscous Force — Frictional force between adjacent layers of a liquid in relative motion
Friction — Force opposing relative motion between surfaces in contact; due to surface irregularities	Static Friction — Friction acting on a body at rest before it starts moving; maximum value = limiting friction
Kinetic Friction — Friction during motion; includes sliding and rolling friction	Sliding Friction — Friction when one surface slides over another surface
Rolling Friction — Friction when a body rolls over a surface; smallest type of friction	Lubricant — Substance that reduces friction by filling gaps between rough surfaces (oil, grease, graphite)
Centre of Gravity — Point through which the total weight of an object appears to act	Stable Equilibrium — State where displaced object returns to original position; CG rises briefly
Unstable Equilibrium — State where displaced object falls away and does NOT return; CG lowers	Neutral Equilibrium — State where displaced object stays in new position; CG height unchanged
Capillarity — Rise of liquid in a narrow tube due to surface tension and adhesion	Knot (unit) — Unit of speed: $1 \text{ knot} = 1 \text{ nautical mile per hour}$; used in aviation and shipping

